

Complex Systems Excercise 4

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1 Fokker - Planck Equation for an Ornstein - Uhlenbeck Process

In this section we start from the Langevin equation for the Ornstein - Uhlenbeck Process:

$$\frac{dv}{dt} = -\frac{v}{\tau} + \xi \quad (1)$$

With

$$\langle \xi(t)\xi(t') \rangle = A\delta(t - t') \quad (2)$$

1.1 The Fokker - Planck

Using the general form of a Langevin equation:

$$\frac{dv}{dt} = -\mu F(v) + \xi \quad (3)$$

We can write:

$$F = v \quad \mu = \tau^{-1} \quad (4)$$

And so the Fokker-Planck equation comes from the continuity equation, where the probability density current is a result of the force and of diffusion:

$$\frac{dp}{dt} = \mu \partial_v [F(v)p] + \partial_v [D \partial_v p] \quad (5)$$

We can set the value for D based on the $\mu = 0$ limit. At that limit, for the delta noise:

$$D = A/2 \quad (6)$$

Plugging this, and the values for μ and $F(v)$:

$$\frac{dp}{dt} = \frac{1}{\tau} \partial_v [pv] + \partial_v \left[\frac{A}{2} \partial_v p \right] \quad (7)$$

We get the Fokker - Planck equation

$$\boxed{\frac{dp}{dt} = \frac{p + vp'}{\tau} + \frac{A}{2} p''} \quad (8)$$

1.2 The velocity at time t

the Langevin equation is just a inhomogenous ODE. The homogenous solution for $\xi = 0$ is:

$$v_h = e^{-t/\tau} \quad (9)$$

We can find the private solution by variation of parameters. Setting:

$$v_p = C(t)e^{-t/\tau} \quad (10)$$

And plugging this into the equation, we get:

$$\left(\dot{C} - \frac{1}{\tau} C \right) e^{-t/\tau} = -\frac{C e^{-t/\tau}}{\tau} + \xi \quad (11)$$

$$\dot{C} = \xi e^{t/\tau} \quad (12)$$

$$c = \int_0^t \xi(t') e^{t'/\tau} dt' \quad (13)$$

In total, we get:

$$v(t) = v_0 e^{-t/\tau} + \int_0^t e^{-(t-t')/\tau} \xi(t') dt' \quad (14)$$

Now, since ξ is a gaussian, and the integral is a linear operator, the final $v(t)$ is also a gaussian. To find it, we need to only find the first and second moments. To do so, we will use the linearity of moments and the known moments of ξ :

$$\mu_\xi = \langle \xi \rangle = 0 \quad \sigma_\xi^2 = \langle \xi^2 \rangle - \langle \xi \rangle^2 = A \quad (15)$$

The first moment:

$$\begin{aligned}
\mu_v &= \langle v_0 e^{-t/\tau} \rangle + \int_0^t e^{-(t-t')/\tau} \langle \xi(t') \rangle dt' = v_0 e^{-t/\tau} \\
\sigma_v^2 &= \langle v^2 \rangle - \langle v \rangle^2 = \\
&= v_0^2 e^{-2t/\tau} + \int_0^t e^{-(t-t')/\tau} \langle v_0 e^{-t'/\tau} \xi(t') \rangle dt' + \\
&\quad + \int_0^t e^{-2(t-t')/\tau} \langle \xi(t') \xi(t') \rangle dt' - v_0^2 e^{-2t/\tau} = \\
&= v_0 e^{-t/\tau} \int_0^t e^{-(t-t')/\tau} \langle \xi(t') \rangle dt' + A \int_0^t e^{-2(t-t')/\tau} dt' = \frac{A\tau}{2} e^{2(t'-t)/\tau} \Big|_{t'=0}^t = \\
&= \frac{A\tau}{2} (1 - e^{-2t/\tau})
\end{aligned} \tag{16}$$

So, this gives the velocity:

$$p(v; t) = \frac{1}{\sqrt{2\pi} \frac{A\tau}{2} (1 - e^{-2t/\tau})} \exp \left[-\frac{1}{2} \left(\frac{v - v_0 e^{-t/\tau}}{\frac{A\tau}{2} (1 - e^{-2t/\tau})} \right)^2 \right] \tag{17}$$

$$p(v; t) = \frac{\sqrt{2}}{\sqrt{\pi} A\tau (1 - e^{-2t/\tau})} \exp \left[-2 \left(\frac{v - v_0 e^{-t/\tau}}{A\tau (1 - e^{-2t/\tau})} \right)^2 \right] \tag{18}$$

It's clear to see that at $t \rightarrow 0$ the distribution is a delta function around $v = v_0$, as expected.

1.3 The $t \rightarrow \infty$ limit

When taking $t \rightarrow \infty$, we get that all the exponents vanish:

$$p(v; t \rightarrow \infty) = \frac{\sqrt{2}}{\sqrt{\pi} A\tau} \exp \left[-2 \left(\frac{v}{A\tau} \right)^2 \right] \tag{19}$$

Which can be understood as the system forgetting the starting conditions and evolving to a steady state gaussian around $v = 0$.

1.4 Thermodynamical interpretation

Considering the steady state as the equilibrium of the system, we can expect the final distribution to be Boltzman distribution:

$$p \propto e^{-H/kT} \tag{20}$$

Comparing this to the final state we got and using the classical kinetic energy:

$$2 \left(\frac{v}{A\tau} \right)^2 = \frac{mv^2}{2kT} \quad (21)$$

Which gives:

$$A = \sqrt{\frac{4kT}{m\tau^2}} \quad (22)$$

This means that the diffusion rate is related linearly to the temperature.